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SHORT COMMUNICATION

Sustainable 3D Printing of Polyhydroxyalkanoates: Material Characterization, Printability, and Parametric Optimization for Enhanced Dimensional Accuracy

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Abstract
This study explores the utilization of Polyhydroxyalkanoates, a bio sourced biodegradable polymer for sustainable additive manufacturing via FDM. The research covers material characterization, printability assessment, and dimensional accuracy optimization. Various characterization techniques including FTIR, DTA, SEM, EDS, and XRD are employed for comprehensive material characterization. Printability assessment identifies challenges with bed adhesion and warping, attributed to PHB's high crystallinity and also helped in determining the processing window by suggesting the range of printing parameters. Parametric optimization using response surface-based desirability identifies optimal sets of parameters for highest dimensional accuracy in all three axes. Crystallinity increased from 40.9% for the as-received filament to 61.8% for the printed parts. Optimal parameters (layer thickness 0.12 mm, extruder temperature 204 °C, flow 100%, print speed 48 mm/sec) yielded dimensional deviations < 0.1 mm from the 15 mm target in all axes with composite desirability of 0.887. Moreover, the confirmatory prints validated prediction error of < 5%.

Keywords 3D printing · Additive manufacturing · Fused deposition modelling · Polyhydroxyalkanoates · Biomaterials · Sustainability

Introduction

Additive manufacturing (or 3D printing) enables the fabrication of intricate geometries through layer-by-layer material deposition [1, 2]. Fused deposition modeling (FDM), a common 3D printing method, offers scalability, affordability, and material versatility [3, 4]. Despite its advantages of waste reduction, traditional plastics used in 3D printing contribute to environmental pollution. Biopolymers, derived from renewable sources like plants and microbes, present sustainable alternatives to petroleum-based plastics [5]. Recent research findings have received significant interest in biopolymers obtained through microbial fermentation like Polyhydroxyalkanoates (PHA). This class of polymer have advantage of lower glass transition temperature, complete biodegradability, thermoplastic, UV resistant, and moreover it leaves no microplastics [6]. Homopolymers of PHA are polyhydroxybutyrate (PHB), polyhydroxyvalerate (PHV), polyhydroxyhexanoate (PHHx), and polyhydroxyoctanoate (PHO) and copolymers as poly(3-hydroxybutyrate-co-3-hydroxyvalerate) (PHBV), poly(3-hydroxybutyrate-co-4-hydroxybutyrate) (P(3HB-co-4HB)), poly(3-hydroxybutyrate-co-3-hydroxyhexanoate) (P(3HB-co-3HHx)), and poly(3-hydroxybutyrate-co-3-hydroxypropionate) (PHBHP) [7]. Among them, PHB is more preferred for AM due to its ease of processability, superior mechanical properties, strong adhesion, and regulatory approval for food contact and medical devices [8]. Compared with PLA (the most common biodegradable FDM filament) and its

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various bio-composites [9–12], neat PHB without any modification offers complete biodegradability across a wider range of environments and can undergo microbial degradation without generating persistent microplastic residues. However, its high crystallinity leads to greater warping and poorer bed adhesion than PLA or ABS, necessitating the detailed printability assessment and RSM optimization presented here.

Processing PHB through FDM presents challenges such as lower printability, higher warpage, and low dimensional accuracy. Despite efforts to improve printability with secondary material blends, it reduces biodegradability. This research identifies printability issues and recommends process settings for successful prints of commercially available PHA filament. A detailed study on the effect of printing process parameters on dimensional accuracy is conducted using response surface methodology (RSM), ANOVA, and pareto analysis for three different axes (X, Y, and Z). A predictive model is obtained and multi-objective optimization achieved using desirability functional analysis (DFA).

Materials and Method

The commercial polyhydroxyalkanoate (PHA) filament was purchased from ColorFabb B.V., Netherlands. It was characterized using FTIR, DTA, FE-SEM, EDS, and XRD to evaluate its chemical structure, thermal behavior, surface morphology, elemental composition, and crystallinity, and was identified as a poly(3-hydroxybutyrate) (PHB) homopolymer grade; however, the exact molecular weight and the presence of any minor copolymer constituents remain proprietary information and were not disclosed by the supplier. The infrared spectra were collected using a Bruker Lumos II FTIR spectrometer in ATR mode. The Perkin-Elmer Pyris Diamond Thermal Analyser is used for DTA, with 10 mg of the sample under an N_2 environment (100 ml/min) from a 26°C to 650°C temperature range and a heating rate of 10°C/min against an alumina reference. FE-SEM and EDS of the cryo-fractured and Au/Pd-coated filament are carried out using the Lyra 3 Tescan system. XRD has been conducted utilizing the PANalytical Empyrean with Cu-Kα radiation at a rate of 2°/min to study the difference in crystallinity between the filament and 3D-printed sample as per S3.

Various geometries are designed using the commercial CAD software (SolidWorks) and then sliced using Ultimaker Cura and finally 3D printed using a direct-drive FDM setup, as shown in Fig. 2a (Creality Ender 3 S1 Pro). Printability assessments are conducted using various geometries (Fig. 2c and d) and varying printing parameters broadly to determine a narrow range providing the best printing results. This range of parameters is further utilized for dimensional accuracy (DA) measurements for a cube sample as shown in Fig. 2e. The experimental design involves four independent input process parameters selected using the Box Behnken method of RSM based DoE, i.e. Layer thickness (A) from 0.08 to 0.20 mm, Extruder temperature (B) from 190 to 210 °C, Flow (C) from 100 to 130%, and extruder speed (D) from 30 to 50 mm/sec for parametric evaluation and its optimization.

Results and Discussion

Characterization of the PHA Filament

ATR-FTIR spectroscopy (Fig. 1d) revealed distinct absorption peaks at 1723 cm^{-1} and 1277 cm^{-1}, attributed to carbonyl functional group stretching vibrations in PHA polymer backbone and alkyl groups within molecular structure respectively. [13, 14]. The FTIR analysis reveals the presence of C-H alkanes group at 2930 cm^{-1} and C-O bonding at 1010 cm^{-1} to 1300 cm^{-1}. The functional groups show uniformity and similarity to PHB, a homopolymer of PHA, confirming its presence [15–17]. The DTA curve exhibited distinctive endothermic dips at specific temperatures (Fig. 1e). The first endothermic dip at 176°C indicates energy absorption, indicating polymer melting, while a prominent dip at 284°C suggests substantial energy absorption, possibly indicating polymer degradation. The surface morphology of PHB filament (Fig. 1a and b) shows triangular and circular porosities due to trapped air bubbles during filament manufacturing. SEM analysis reveals filament dimensions ranging from 1710 µm to 1730 µm, contrary to the manufacturer's stated 1750 µm. The EDS analysis (Fig. 1c) reveals carbon and oxygen as the primary biopolymer constituents, with minor peaks from Au/Pd coating and traces of Mg and Si detected. The normalised quantitative analysis shows a 43.1% carbon weight percentage and 56.9% oxygen weight percentage. The XRD analysis reveals that the main peaks at Bragg angles of 13.61°, 16.9°, 19.2°, 22.2°, 25.5°, 26.9°, and 28.8° match well with previous literature [18] and ICDD 00-049-2212 data. Other peaks at 37.8°, 44.1°, 48.9°, 64.20°, 70.35°, and 77.27° are observed mainly for the 3D-printed sample, depicting an increased crystallinity. Quantitative analysis of XRD data (using the method in Supplementary Data S2) yielded a crystallinity of approximately 40.9% for the as-received filament. After FDM processing, crystallinity increased to around 61.8%. This rise is attributed to flow-induced orientation and shear crystallization during extrusion, followed by relatively slow cooling in ambient conditions, allowing more time for chain folding and lamellae growth in the highly stereoregular PHB.

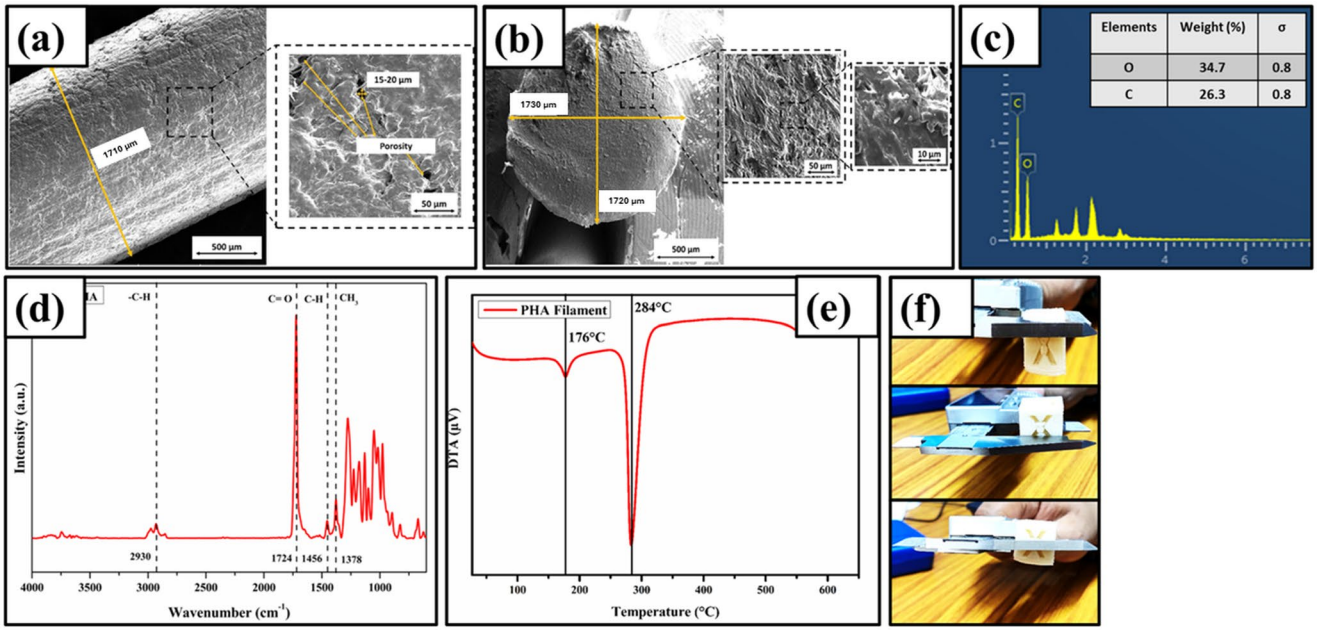


Fig. 1 Characterization of PHB filament **a** FE-SEM morphology of the filament side surface, **b** FE-SEM morphology of the cross section, **c** EDS Analysis, **d** ATR-FTIR, **e** DTA analysis, **f** Dimension measurement at three different positions for three different axes

Printability Assessment

The adhesion of printed PHB to the print bed was challenging, so various support methods were explored. The brim, with its extended surface area, was found to reduce warping and enhance adhesion. Bed temperature significantly impacts warping and bed adhesion in FDM printing. However, PHB has higher crystallinity due to its stereoregular crystalline structure [19], which can lead to significant thermal contraction when cooling down after extrusion. The optimal condition is no bed heating and printing PHB filament in ambient conditions (27–30 °C with humidity around 50–70%). The optimal extruder temperature was found to be 190–210 °C and the optimum print speed range was 30–50 mm/sec from the testing towers (Fig. 2c). Warping was explored using geometries (Supplementary data (S3)) with limited contact points, and a direct correlation was found between processing parameters and warping.

Parametric Optimization of Dimensional Accuracy

Deviations of the printed part from the intended CAD geometry can arise from heat deformation, material shrinkage, and layer misalignment, especially for a difficult-to-print material such as PHB. The normal probability plot is utilized to assess the distribution of residuals, and they adhered to a normal pattern as shown in Fig. 2f. The regression model's robustness is demonstrated by high R^2 values across all measured directions. The adjusted R^2 values also align with predicted R^2, validating the model's higher predictability with low error. The regression equations are presented in equations S2.1, S2.2, and S2.3 (Supplementary data).

Effect of Process Parameters on DA

ANOVA (Supplementary Table S1.2) and Pareto analysis (Fig. 2f) demonstrate that material flow percentage (factor C) exerts by far the strongest main effect on dimensional accuracy in all three axes, with extremely high F-values of 663.1 (X), 483.0 (Y), and 704.8 (Z) (all $p<0.0001$). This dominance is physically expected as flow rate directly governs the volume of molten PHB extruded per unit length, thereby controlling the deposited bead width, height, and the compressive pressure at bead-to-bead and layer-to-layer interfaces. Higher flow promotes better intimate contact and molecular chain diffusion across interfaces during the short time, as the material remains above T_g, reducing voids and differential shrinkage stresses that otherwise distort the part. The statistically significant AC interaction further reveals that the benefit of increased flow is strongly modulated by layer thickness. At lower layer thicknesses, elevated flow improves X and Y directional accuracy through enhanced lateral bonding and reduced in-plane shrinkage; however, the same combination can slightly degrade Z-accuracy because excessive material volume per layer increases cumulative stacking error and amplifies the effect of PHB's high crystallinity-driven volumetric contraction along the build direction.

Moreover, Run 1 has the best DA in the Z direction, while Run 25 has the best results in the X and Y directions

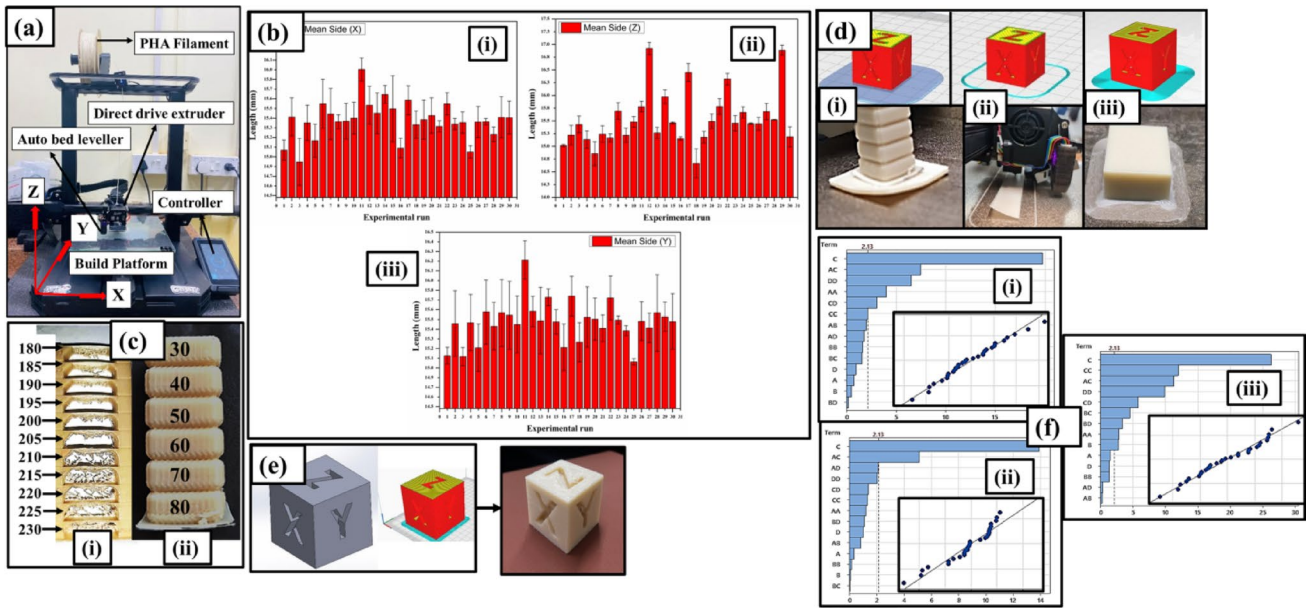


Fig. 2 Fabrication and Results **a** FDM 3D printing setup, **b** Dimensional accuracy results for each run based on DoE for (i) X direction (ii) Z direction, and (iii)Y direction, **c** Test specimen for printability in (i) Temperature tower (ii) Speed tower, **d** Built plate adhesion types for (i) Raft (ii) Skirt (iii) Brim, **e** Sample for dimensional accuracy measurement, **f** Pareto chart and Normal probability plot (inserted figure) for (i) X direction (ii) Y direction, and (iii)Z direction

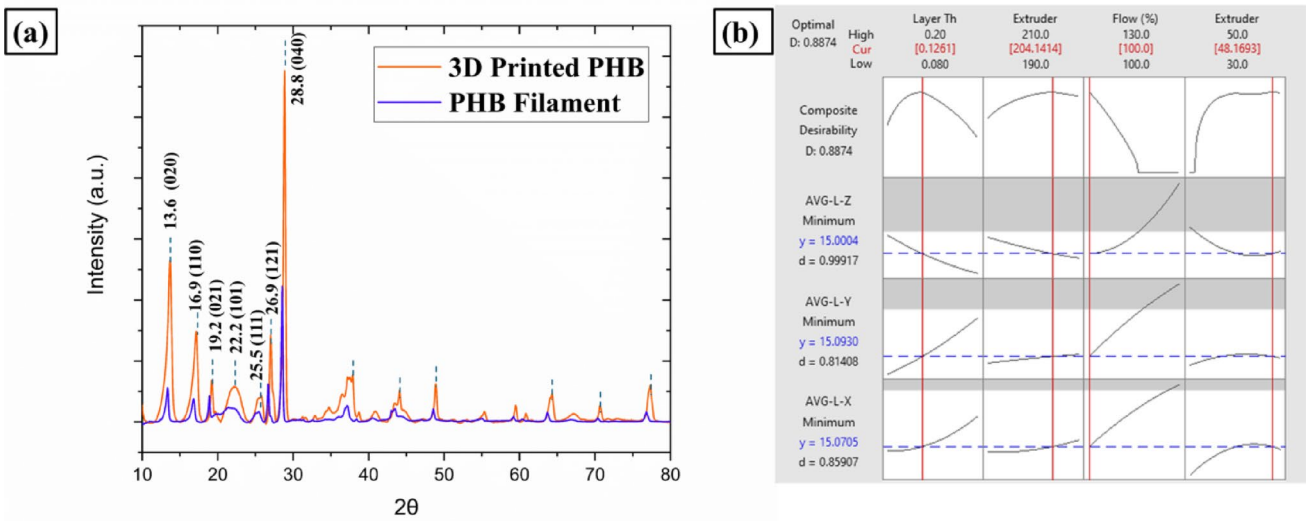


Fig. 3 **a** XRD result for PHB filament and its 3D printed part **b** Multi objective optimization of DA for all three axes using DFA

(Table S1.1). The most important process parameters affecting Z direction DA are flow and extruder temperature. . A lower layer thickness improves dimensional accuracy in X and Y directions, but may hinder Z direction accuracy due to layer bonding issues. Higher flow rate enhances material flow and layer adhesion, but Z direction accuracy remains unimproved.

Parametric Optimization Using DFA

A multi-objective optimization is conducted using DFA to get the best dimensional accuracy in X, Y, and Z directions concurrently. Among many solutions obtained, the one with highest composite desirability of around 88.7% is selected for the confirmatory test as shown in Fig. 3b. The quadratic models exhibited high adequacy with $R^2 > 0.95$ and close agreement between adjusted and predicted R^2 for all three axes as shown in Fig. 4 (full statistics in Supplementary

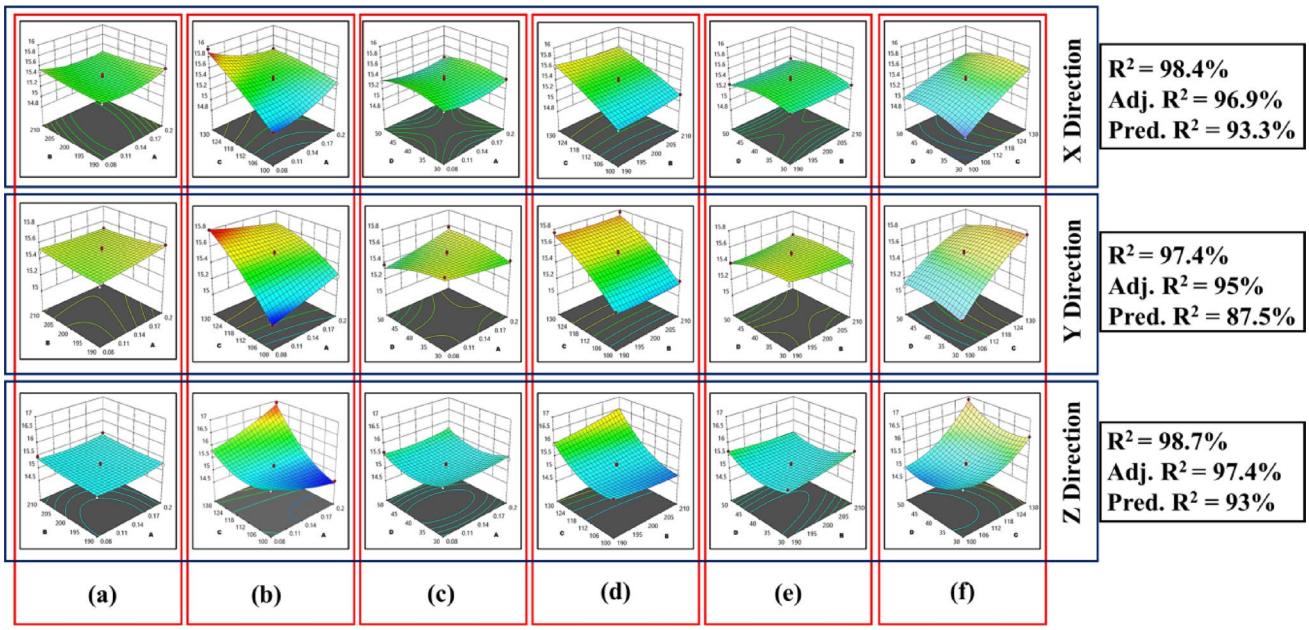


Fig. 4 3D Surface plot of the interaction between various process parameters and the length measured in X, Y, and Z direction where **a** A: Layer thickness vs B: Extruder temperature, **b** A: Layer thickness vs C: Flow, **c** A: Layer thickness vs D: Extruder speed, **d** B: Extruder temperature vs C: Flow, **e** B: Extruder temperature vs D: Extruder speed, **f** C: Flow vs D: Extruder speed

Table S1.2). Lack-of-fit was insignificant ($p > 0.05$). Confirmatory prints at the optimal solution (A = 0.12 mm, B = 204 °C, C = 100%, D = 48 mm/sec) produced mean lengths of 15.00–15.09 mm, confirming prediction error < 5%. Compared with recently reported PLA [20, 21] and other PHB blends/composites [22], the present work demonstrates that high dimensional accuracy can be achieved with unmodified commercial PHB through targeted parametric optimization alone, without blending or additives that may compromise full biodegradability. This is particularly relevant for applications requiring both sustainability and regulatory compliance. Future studies should focus on unlocking the full potential of PHB for sustainable 3D, 4D [23], and 5D printing [24], spanning macro, micro, and nanoscale fabrication [25]. Such efforts could facilitate the development of geometrically complex, functionally graded, and stimuli-responsive biodegradable structures [26] capable of addressing emerging challenges in biomedical devices, soft robotics, smart materials, tissue engineering, and environmentally adaptive systems, thereby broadening the scope of eco-friendly advanced manufacturing.

Conclusion

This study characterized a commercially available PHA-based filament and its FDM-printed parts and established a practical processing window for FDM. A four-factor Box–Behnken design and response surface methodology were employed to evaluate dimensional accuracy along the X, Y, and Z directions. Multi-objective optimization identified implementable processing conditions of 0.12 mm layer thickness, 204 °C extruder temperature, 100% flow, and 48 mm/s print speed, producing dimensional deviations below 0.1 mm from the nominal 15 mm target in all three directions. Achieving sub-0.1 mm deviation from the target geometry across all axes is critical for functional FDM parts, as even small dimensional errors can accumulate in assemblies, affect load distribution in lattice structures, or compromise fit in patient-specific biomedical devices. The optimized parameters identified here, therefore, offer direct translational value for sustainable manufacturing of high-accuracy PHB components.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40009-026-02354-w.

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